ZAINAB YOUSAF

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PROFESSIONAL SUMMARY

AI Engineer who builds production AI systems that work under real load not just demos. Built **DreamDesk**, a multi-agent platform that autonomously handles 100% of hotel guest communications across email, chat, and WhatsApp with zero dedicated support staff. Research background includes collaboration with **DFKI Germany** on smart contract security. Specialized in multi-agent LLM pipelines, RAG architectures, and end-to-end SaaS development using Pydantic-AI, FastAPI, Pinecone, Next.js, and NestJS. Known for clean architecture, evaluation-driven development, and shipping things that scale.

EDUCATION

FAST-NUCES — **Faisalabad, Pakistan**
BS Computer Science | CGPA: 3.34 — *Sep 2020 – Jun 2024*

- Dean's List (2x) awarded for outstanding academic performance in consecutive semesters.

TECHNICAL SKILLS

AI & LLM: Pydantic-AI, LangChain, Pinecone (RAG), OpenAI API, OpenRouter, Google Gemini 2.5 Flash, MCP Tool-Calling, OpenAI Embeddings, Prompt Engineering, Firecrawl, PyMuPDF
Auth & Integrations: Clerk, Svix (Webhooks), Microsoft Graph SDK, Google APIs, Mews PMS SDK, Amazon Seller API
Frontend: React.js, Next.js, TailwindCSS, TanStack Query, React Flow, Recharts, Framer Motion, TypeScript, JavaScript, HTML, CSS
Backend: FastAPI, NestJS, Node.js, Express.js, SQLAlchemy, Alembic, REST APIs, WebSockets, Microservices, PostgreSQL, MongoDB, MySQL, Redis
Observability & Evals: Logfire, PostHog, Confidence Scoring, Auto-Escalation Pipelines
Mobile: Swift, SwiftUI, AppKit (macOS), UIKit, Swift Concurrency, VisionKit
Cloud & Tools: Google Cloud SQL, AWS, Docker, Git, Postman, Jira, Pandas, NumPy, OpenCV

WORK EXPERIENCE

AI Engineer — **Python, FastAPI, Next.js, TypeScript, Pydantic-AI**
DreamDesk — *Jan 2026 – Present*

- Built **DreamDesk** from the ground up a production multi-agent platform that autonomously handles 100% of hotel guest inquiries across email, chat, and WhatsApp, replacing dedicated per-channel support staff entirely.
- Architected a 4-layer Pydantic-AI agent graph (*understand* → *plan* → *execute* → *commit*) with zero-LLM-cost routing and confidence-based auto-escalation to reduce hallucination risk in live guest communications.
- Integrated Pinecone RAG with Gemini 2.5 Flash for semantic retrieval, Mews PMS SDK for live reservation management, and MCP tool-calling connecting agents to Microsoft Graph, Gmail, and Google Maps on Google Cloud SQL.
- Delivered multi-tenant Next.js dashboard with React Flow workflow builder and Clerk auth; wired PostHog for product analytics and Logfire for full observability and tracing.

AI Engineer — **NestJS, React.js, TypeScript, PostgreSQL, OpenAI API**
AutoMato AI — *Jul 2025 – Dec 2025*

- Built an OpenAI GPT-powered pipeline helping 50+ multinational companies automate Amazon product listings, reducing manual listing effort by 40–60% through prompt engineering and SEO-optimized retrieval.
- Integrated Amazon Seller API to pull live competitor data and extract high-ranking keywords; architected NestJS microservices backend with PostgreSQL and React.js dashboard for end-to-end listing review and publishing.

Software Engineer - iOS/macOS — **Swift, SwiftUI, UIKit, AppKit, Swift Concurrency**
KATco — *Jan 2025 – Jul 2025*

- Engineered and shipped **PDF Converter** (iOS/macOS, App Store) and **BGZoom** (professional macOS video-calling app with real-time background effects), integrating Cloud Converter API and optimizing with Swift Concurrency.
- Reduced manual processing effort by 40% through API integration and improved processing speed by 30% with Swift Concurrency optimizations across both apps.

PROJECTS

DreamDesk: Multi-Channel AI Agent Pipeline — **Pydantic-AI, MCP, LangChain, Pinecone, Firecrawl, PyMuPDF**
AI/Backend Architecture — *Jan 2026 – Present*

- Designed the core agent graph with 4 structured Pydantic-AI layers, enforcing a single DB write boundary at commit to guarantee data consistency across all guest communication channels.
- Built a knowledge ingestion pipeline using Firecrawl (web scraping) and PyMuPDF (PDF parsing) to populate the hotel knowledge hub, indexed into Pinecone for low-latency semantic retrieval.
- Implemented MCP tool-calling connecting agents to Microsoft Graph, Gmail, Google Maps, and Svix webhooks; integrated PostHog for product analytics and Logfire for full observability and tracing.

AutoMato AI: Amazon Listing Intelligence (Link) — **NestJS, React.js, Amazon Seller API, OpenAI API**
Full Stack AI Development — *Jul 2025 – Dec 2025*

- Integrated Amazon Seller API to pull live competitor data; built an OpenAI-powered pipeline to extract high-ranking keywords and generate SEO-optimized listing content automatically.
- Architected NestJS microservices backend with PostgreSQL; designed React.js dashboard enabling sellers to review, edit, and publish AI-generated listings in a single workflow.

BGZoom: Professional macOS Video Calling App — **Swift, SwiftUI, AppKit, Swift Concurrency**
Associated with KATco — *Mar 2025 – Jun 2025*

- Developed **BGZoom**, a professional macOS video-calling app supporting Zoom, Google Meet, Skype, and Teams with custom real-time background effects and native AppKit UI.
- Implemented high-performance video/audio streaming using Swift Concurrency, ensuring smooth real-time collaboration with low latency across platforms.

PDF Converter: Photo to PDF (App Store) — **Swift, SwiftUI, UIKit, Cloud Converter API**
Published iOS/macOS App

- Published iOS/macOS app on the App Store enabling seamless file format conversions into high-quality PDFs; improved processing speed by 30% and reduced system workload by 40% via Swift Concurrency optimizations.

RESEARCH

Information Flow Guided Grey-Box Fuzzing — **Solidity, Echidna, Foundry, Python, Slither**
Research Collaboration DFKI Germany & FAST-NUCES — *Aug 2023 – Apr 2024*

- Designed a novel technique combining static analysis with grey-box fuzzing to maximize smart contract vulnerability detection while minimizing computational overhead reducing resource usage by 25%.
- Collaborated with researchers from DFKI Germany and NUCES Pakistan to validate results, contributing to decentralized application security research across two institutions.